

LECTURES ON ALGEBRAIC GEOMETRY

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1 Lecture 1: August 10

Throughout this lecture, k is an algebraically closed field, for example $k = \mathbb{C}$. We write $\mathbb{A}^n = \mathbb{A}_k^n$ for affine n -space over k .

1.1 Affine varieties and Zariski topology

Definition 1.1. Let $S \subseteq k[x_1, \dots, x_n]$. Its vanishing locus is

$$V(S) = \{P \in \mathbb{A}^n : f(P) = 0 \text{ for every } f \in S\}.$$

A subset of $\mathbb{A}^n$ of the form $V(S)$ is called an *affine algebraic set*, or, following the terminology of the lecture, an *affine variety*. When irreducibility is required, it will be stated explicitly.

The Zariski topology on $\mathbb{A}^n$ is the topology whose closed subsets are the affine algebraic sets. If $X = V(S) \subseteq \mathbb{A}^n$, then X is given the induced Zariski topology.

We now repackage these geometric objects in algebraic language.

Definition 1.2. Let $X = V(S) \subseteq \mathbb{A}^n$. The *ideal* of X is

$$I(X) = \{f \in k[x_1, \dots, x_n] : f(x) = 0 \text{ for every } x \in X\}.$$

The *coordinate ring* of X is

$$A(X) = k[x_1, \dots, x_n]/I(X).$$

Elements of $A(X)$ are precisely the polynomial functions on X .

The ideal $I(X)$ is radical. More precisely, Hilbert's Nullstellensatz gives

$$I(V(S)) = \sqrt{(S)},$$

where (S) denotes the ideal generated by S . Thus $I(X)$ should not in general be confused with (S) itself. Moreover, X is irreducible if and only if $I(X)$ is prime, or equivalently, if and only if $A(X)$ is an integral domain.

1.2 Regular functions

Next, let us take a closer look at what happens inside the Zariski topology.

Definition 1.3. Let X be an affine variety and let $U \subseteq X$ be open. A map $\phi: U \rightarrow k$ is a *regular function* if, for every $a \in U$, there exist an open neighbourhood $U_a \subseteq U$ of a and elements $f_a, g_a \in A(X)$ such that f_a is nowhere zero on U_a and

$$\phi(x) = \frac{g_a(x)}{f_a(x)} \quad \text{for every } x \in U_a.$$

The set of all regular functions on U is denoted by $\mathcal{O}_X(U)$. Under pointwise addition and multiplication, it is a k -algebra.

The word “locally” in this definition is essential: different quotients may be used on different open subsets, provided that they agree on the overlaps.

Example 1.4. Let

$$X = V(x_1x_4 - x_2x_3) \subseteq \mathbb{A}^4$$

and let

$$U = X \setminus V(x_2, x_4).$$

Define $\phi: U \rightarrow k$ by

$$\phi(x_1, x_2, x_3, x_4) = \begin{cases} \frac{x_1}{x_2}, & x_2 \neq 0, \\ \frac{x_3}{x_4}, & x_4 \neq 0. \end{cases}$$

This is well-defined wherever both x_2 and x_4 are nonzero because

$$x_1x_4 - x_2x_3 = 0 \implies \frac{x_1}{x_2} = \frac{x_3}{x_4}.$$

Hence $\phi \in \mathcal{O}_X(U)$. This example illustrates why a regular function is defined by local quotients; one should not expect an arbitrary open set to admit a single preferred denominator.

Lemma 1.5. *Let $U \subseteq X$ be open and let $\phi \in \mathcal{O}_X(U)$. Then*

$$V_U(\phi) = \{x \in U : \phi(x) = 0\}$$

is closed in U .

Proof. For every $a \in U$, choose an open neighbourhood $U_a \subseteq U$ and $f_a, g_a \in A(X)$ such that

$$\phi(x) = \frac{g_a(x)}{f_a(x)} \quad (x \in U_a),$$

where f_a has no zero on U_a . It follows that

$$U_a \setminus V_U(\phi) = U_a \setminus V(g_a),$$

which is open in U_a , and hence open in U . Note that $U = \bigcup_{a \in U} U_a$. Therefore

$$U \setminus V_U(\phi) = \bigcup_{a \in U} (U_a \setminus V_U(\phi))$$

is open in U , so $V_U(\phi)$ is closed in U . □

Corollary 1.6 (Identity principle). *Assume that X is irreducible. Let $U \subseteq V \subseteq X$ be open subsets with $U \neq \emptyset$. If $\phi_1, \phi_2 \in \mathcal{O}_X(V)$ and $\phi_1 = \phi_2$ on U , then $\phi_1 = \phi_2$ on V .*

Proof. Apply Lemma 1.5 to $\phi_1 - \phi_2$. Its zero locus is closed in V and contains the nonempty open subset U . Since X is irreducible, U is dense in X , and hence dense in V . Therefore the zero locus is all of V . □

Remark (Analogy with complex analysis). This is the algebraic-geometric analogue of the identity theorem in complex analysis: two holomorphic functions on a connected domain that agree on a nonempty open subset agree everywhere. Here irreducibility guarantees that every nonempty open subset is dense, which is the key topological ingredient in the argument above.

1.3 Distinguished open subsets

Definition 1.7. For $f \in A(X)$, the *distinguished open subset* associated with f is

$$D(f) = X \setminus V(f) = \{x \in X : f(x) \neq 0\}.$$

These open subsets satisfy

$$D(1) = X, \quad D(fg) = D(f) \cap D(g), \quad D(f) \cup D(g) = D((f, g)).$$

In the last expression, $D((f, g))$ is shorthand for $X \setminus V(f, g)$.

Lemma 1.8. *The distinguished open subsets form a basis for the Zariski topology on X . In particular, if $a \in U \subseteq X$ and U is open, then there is some $h \in A(X)$ such that*

$$a \in D(h) \subseteq U.$$

Proof. Write $X \setminus U = V(J)$ for an ideal $J \subseteq A(X)$. Since $a \notin V(J)$, some $h \in J$ satisfies $h(a) \neq 0$. Then $a \in D(h)$, while $V(J) \subseteq V(h)$ implies $D(h) \subseteq U$. $\square$

Proposition 1.9. *Let X be an irreducible affine variety and let $f \in A(X)$. Then*

$$\mathcal{O}_X(D(f)) = \left\{ \frac{g}{f^n} \Big|_{D(f)} : g \in A(X), n \in \mathbb{Z}_{\geq 0} \right\}.$$

Proof. The inclusion “ $\supseteq$ ” is immediate. We prove “ $\subseteq$ ”. Let $\phi \in \mathcal{O}_X(D(f))$. For every $a \in D(f)$, there is an open neighbourhood $U_a \subseteq D(f)$ such that

$$\phi(x) = \frac{g_a(x)}{f_a(x)} \quad (x \in U_a),$$

where $f_a, g_a \in A(X)$ and f_a has no zero on U_a .

We make the following two reductions:

- By Lemma 1.8, shrink U_a so that $U_a = D(h_a)$ for some $h_a \in A(X)$.
 - Replace g_a by $g_a h_a$ and f_a by $f_a h_a$.
- The new denominator is zero on $V(h_a)$ and nonzero on $D(h_a)$. Consequently its distinguished open set is exactly $D(h_a)$.

After renaming the new numerator and denominator, we may therefore assume

$$a \in D(f_a) \subseteq D(f), \quad \phi = \frac{g_a}{f_a} \quad \text{on } D(f_a).$$

On $D(f_a) \cap D(f_b)$, we have

$$\frac{g_a}{f_a} = \frac{g_b}{f_b} \implies g_a f_b - g_b f_a = 0.$$

Because X is irreducible, the intersection of the two nonempty open sets $D(f_a)$ and $D(f_b)$ is nonempty. By Corollary 1.6,

$$g_a f_b - g_b f_a = 0 \quad \text{in } A(X). \tag{1.1}$$

Now

$$D(f) = \bigcup_{a \in D(f)} D(f_a) \implies V(f) = \bigcap_{a \in D(f)} V(f_a) = V((f_a : a \in D(f))).$$

Hilbert’s Nullstellensatz gives

$$f \in \sqrt{(f_a : a \in D(f))}.$$

Hence there are an integer $N \geq 1$, a finite subset $F \subseteq D(f)$, and elements $k_a \in A(X)$ for $a \in F$ such that

$$f^N = \sum_{a \in F} k_a f_a.$$

Take

$$g = \sum_{a \in F} k_a g_a.$$

For any $b \in D(f)$, equation (1.1) yields, on $D(f_b)$,

$$g f_b = \sum_{a \in F} k_a g_a f_b = \sum_{a \in F} k_a g_b f_a = f^N g_b.$$

Therefore

$$\phi = \frac{g_b}{f_b} = \frac{g}{f^N} \quad \text{on } D(f_b).$$

Since the sets $D(f_b)$ cover $D(f)$, the equality holds on all of $D(f)$. $\square$

Corollary 1.10. *Let X be an irreducible affine variety and let $f \in A(X)$. Then*

$$\mathcal{O}_X(D(f)) \cong A(X)_f.$$

Proof. If $f = 0$, then $D(f) = \emptyset$ and both sides are the zero ring. Assume $f \neq 0$. Construct

$$\Theta: A(X)_f \longrightarrow \mathcal{O}_X(D(f)), \quad \frac{g}{f^n} \longmapsto \left(x \longmapsto \frac{g(x)}{f(x)^n} \right).$$

We check that Θ is well-defined. Suppose

$$\frac{g}{f^n} = \frac{h}{f^m} \quad \text{in } A(X)_f.$$

By the definition of localization, there exists $r \geq 0$ such that

$$f^r(f^m g - f^n h) = 0 \quad \text{in } A(X).$$

For every $x \in D(f)$, we have $f(x) \neq 0$; evaluating the last identity at x and dividing by $f(x)^{r+m+n}$ gives

$$\frac{g(x)}{f(x)^n} = \frac{h(x)}{f(x)^m}.$$

Thus the resulting regular function is independent of the chosen representative, and Θ is a well-defined k -algebra homomorphism.

Proposition 1.9 shows that Θ is surjective. If $\Theta(g/f^n) = 0$, then g vanishes on the nonempty open subset $D(f)$. Applying Corollary 1.6 to the regular functions g and 0 on X , we obtain $g = 0$ in $A(X)$. Hence $g/f^n = 0$ in $A(X)_f$, so Θ is injective as well. $\square$

In particular, taking $f = 1$ gives $\mathcal{O}_X(X) \cong A(X)$.

2 Lecture 2: August 12

In the previous lecture, regular functions were assigned to every open subset of an affine variety. We now isolate the formal properties of this assignment. This leads to sheaves, stalks, and an intrinsic definition of affine varieties.

2.1 Presheaves and sheaves

Let X be a topological space. If $U \subseteq V$ are open subsets, one should think of a function on V as giving, by restriction, a function on U .

Definition 2.1. A *presheaf of rings* $\mathcal{F}$ on X consists of the following information:

- (1) for every open subset $U \subseteq X$, a ring $\mathcal{F}(U)$;
- (2) for every inclusion of open subsets $U \subseteq V$, a ring homomorphism

$$\rho_{V,U}: \mathcal{F}(V) \longrightarrow \mathcal{F}(U),$$

called the *restriction homomorphism*.

This information is required to satisfy

$$\mathcal{F}(\emptyset) = \{0\}, \quad \rho_{U,U} = \text{id}_{\mathcal{F}(U)}, \quad \rho_{W,U} = \rho_{V,U} \circ \rho_{W,V}$$

whenever $U \subseteq V \subseteq W$ are open. We often write $s|_U$ instead of $\rho_{V,U}(s)$.

An element $s \in \mathcal{F}(U)$ is called a *section* of $\mathcal{F}$ over U .

Thus a presheaf records local information together with a consistent notion of restriction. A sheaf is a presheaf for which compatible local information can be glued uniquely.

Definition 2.2. A presheaf $\mathcal{F}$ on X is a *sheaf* if it satisfies the following *gluing property*. Let $U \subseteq X$ be open, let $U = \bigcup_{i \in I} U_i$ be an open cover, and suppose that $s_i \in \mathcal{F}(U_i)$ satisfy

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \quad \text{for all } i, j \in I.$$

Then there exists a unique section $s \in \mathcal{F}(U)$ such that

$$s|_{U_i} = s_i \quad \text{for every } i \in I.$$

Proposition 2.3 (The most important example). *Let X be an affine variety. The assignment*

$$U \mapsto \mathcal{O}_X(U),$$

together with the usual restriction of functions, is a sheaf of rings on X . It is called the structure sheaf of X .

Proof. The restriction maps clearly satisfy the presheaf axioms. Let $U = \bigcup_{i \in I} U_i$ and suppose that $\phi_i \in \mathcal{O}_X(U_i)$ agree on all pairwise intersections. Define

$$\phi(x) = \phi_i(x) \quad \text{if } x \in U_i.$$

Compatibility on the overlaps shows that ϕ is well-defined. For each $x \in U$, choose i with $x \in U_i$. Near x , the function ϕ agrees with the regular function ϕ_i , so ϕ is regular. It is plainly the unique function with the prescribed restrictions. $\square$

2.2 Stalks and local rings

A sheaf describes functions on every open set. To retain only the behaviour arbitrarily close to one point, we pass to germs.

Definition 2.4. Let $\mathcal{F}$ be a presheaf on X and let $a \in X$. The *stalk* of $\mathcal{F}$ at a is

$$\mathcal{F}_a = \{(U, s) : a \in U \text{ open, } s \in \mathcal{F}(U)\} / \sim,$$

where

$$(U, s) \sim (U', s')$$

if there exists an open neighbourhood $V \subseteq U \cap U'$ of a such that

$$s|_V = s'|_V.$$

The equivalence class of (U, s) is denoted by s_a and is called the *germ* of s at a .

Addition and multiplication are defined after restricting two representatives to a common neighbourhood of a . The presheaf axioms show that these operations are well-defined.

Lemma 2.5. *Let $X \subseteq \mathbb{A}^n$ be an affine variety and let $a \in X$. Write*

$$\mathfrak{m}_a = \{f \in A(X) : f(a) = 0\}$$

for the maximal ideal corresponding to a . Then

$$\mathcal{O}_{X,a} \cong A(X)_{\mathfrak{m}_a}.$$

Proof. Consider the homomorphism

$$\Theta_a : A(X)_{\mathfrak{m}_a} \longrightarrow \mathcal{O}_{X,a}, \quad \frac{g}{f} \longmapsto \left(D(f), \frac{g}{f} \right)_a.$$

Here $f \notin \mathfrak{m}_a$, so $f(a) \neq 0$ and hence $a \in D(f)$.

We first check that Θ_a is well-defined. Suppose

$$\frac{g}{f} = \frac{g'}{f'} \quad \text{in } A(X)_{\mathfrak{m}_a}.$$

Then there exists $h \notin \mathfrak{m}_a$ such that

$$h(gf' - g'f) = 0 \quad \text{in } A(X).$$

The open subset

$$D(h) \cap D(f) \cap D(f')$$

contains a . On this neighbourhood, h , f , and f' are nonzero, and the last identity gives

$$\frac{g}{f} = \frac{g'}{f'}.$$

Thus the two quotients determine the same germ at a .

To prove injectivity, suppose that $\Theta_a(g/f) = 0$. There is an open neighbourhood $V \subseteq D(f)$ of a on which $g/f = 0$. Choose $h \in A(X)$ with

$$a \in D(h) \subseteq V.$$

Then g vanishes on $D(h)$. On $V(h)$ the function h vanishes, so the product hg vanishes everywhere on X . Consequently $hg = 0$ in $A(X)$. Since $h(a) \neq 0$, we have $h \notin \mathfrak{m}_a$, and therefore $g/f = 0$ in $A(X)_{\mathfrak{m}_a}$.

Finally, let a germ in $\mathcal{O}_{X,a}$ be represented by (U, ϕ) , where $a \in U$. By the definition of a regular function, after shrinking around a there are $g, f \in A(X)$ such that

$$\phi = \frac{g}{f}$$

and f is nonzero on that neighbourhood. In particular $f(a) \neq 0$, so $f \notin \mathfrak{m}_a$, and the germ of ϕ is $\Theta_a(g/f)$. Hence Θ_a is surjective. $\square$

The lemma explains why $\mathcal{O}_{X,a}$ is called the *local ring* of X at a : its unique maximal ideal consists of germs of regular functions that vanish at a .

2.3 Ringed spaces and morphisms

Definition 2.6. A *ringed space* is a topological space X together with a sheaf of rings $\mathcal{O}_X$. We denote it by $(X, \mathcal{O}_X)$, or simply by X when the sheaf is understood.

In the geometric setting of this lecture, the sections of $\mathcal{O}_X$ are k -valued functions. If $f: X \rightarrow Y$ is a map, $U \subseteq Y$ is open, and $\phi \in \mathcal{O}_Y(U)$, the pullback is represented by the composite

$$f^{-1}(U) \xrightarrow{f} U \xrightarrow{\phi} k,$$

and is defined by

$$f^*\phi = \phi \circ f: f^{-1}(U) \rightarrow k.$$

Definition 2.7. A map $f: X \rightarrow Y$ between ringed spaces of functions is called a *morphism* if

- (1) f is continuous;
- (2) for every open subset $U \subseteq Y$ and every $\phi \in \mathcal{O}_Y(U)$, one has

$$f^*\phi \in \mathcal{O}_X(f^{-1}(U)).$$

An *isomorphism* is a morphism that admits a morphism as its inverse.

Condition (2) says that pulling back regular functions produces regular functions. In the general language of ringed spaces, these pullback maps form the sheaf homomorphism that is part of the morphism.

Proposition 2.8. *Let $X \subseteq \mathbb{A}^m$ and $Y \subseteq \mathbb{A}^n$ be affine varieties, and let $U \subseteq X$ be open. The morphisms $f: U \rightarrow Y$ are exactly the maps of the form*

$$f(x) = (\phi_1(x), \dots, \phi_n(x)), \quad \phi_1, \dots, \phi_n \in \mathcal{O}_X(U),$$

whose values lie in Y .

Proof. Suppose first that f is a morphism. Let $y_1, \dots, y_n \in \mathcal{O}_Y(Y)$ be the coordinate functions on Y . By definition,

$$\phi_i = f^* y_i \in \mathcal{O}_X(U) \quad (1 \leq i \leq n),$$

and therefore

$$f(x) = (\phi_1(x), \dots, \phi_n(x)).$$

Conversely, suppose that f has this form. We first prove continuity. Every closed subset $Z \subseteq Y$ can be written as

$$Z = Y \cap V(g_1, \dots, g_r)$$

for some $g_1, \dots, g_r \in k[y_1, \dots, y_n]$. Hence

$$f^{-1}(Z) = \{x \in U : g_j(\phi_1(x), \dots, \phi_n(x)) = 0 \text{ for every } j\}.$$

Each $g_j(\phi_1, \dots, \phi_n)$ is regular on U , so its zero locus is closed in U by Lemma 1.5. Thus $f^{-1}(Z)$ is closed and f is continuous.

It remains to check pullbacks of regular functions. Let $W \subseteq Y$ be open, let $\psi \in \mathcal{O}_Y(W)$, and take $x \in f^{-1}(W)$. On a neighbourhood of $f(x)$ in W , write

$$\psi = \frac{p}{q}$$

with $p, q \in A(Y)$ and q nowhere zero there. On the corresponding neighbourhood of x ,

$$f^* \psi = \frac{p(\phi_1, \dots, \phi_n)}{q(\phi_1, \dots, \phi_n)},$$

and the denominator is nowhere zero. Therefore $f^* \psi$ is regular, and f is a morphism. $\square$

Corollary 2.9. *For affine varieties $X \subseteq \mathbb{A}^m$ and $Y \subseteq \mathbb{A}^n$, pullback gives a natural bijection*

$$\text{Mor}(X, Y) \xrightarrow{\sim} \text{Hom}_{k\text{-alg}}(A(Y), A(X)), \quad f \mapsto f^*.$$

Thus morphisms of affine varieties correspond contravariantly to homomorphisms of their coordinate rings.

Proof. Every morphism $f: X \rightarrow Y$ induces the k -algebra homomorphism

$$f^*: A(Y) = \mathcal{O}_Y(Y) \longrightarrow \mathcal{O}_X(X) = A(X), \quad \psi \mapsto \psi \circ f.$$

Conversely, let

$$\alpha: A(Y) \longrightarrow A(X)$$

be a k -algebra homomorphism, and let $\bar{y}_1, \dots, \bar{y}_n$ denote the coordinate classes in $A(Y)$. Define

$$f_\alpha: X \longrightarrow \mathbb{A}^n, \quad a \mapsto (\alpha(\bar{y}_1)(a), \dots, \alpha(\bar{y}_n)(a)).$$

If $g \in I(Y)$, then $g(\bar{y}_1, \dots, \bar{y}_n) = 0$ in $A(Y)$. Applying α shows that g vanishes at every point $f_\alpha(a)$. Hence $f_\alpha(X) \subseteq Y$, and Proposition 2.8 shows that $f_\alpha: X \rightarrow Y$ is a morphism.

The two constructions are inverse. Indeed, the coordinate functions of the map recovered from f^* are

$$f^*(\bar{y}_i) = \bar{y}_i \circ f,$$

so the recovered map is f . Conversely, for every polynomial class $\bar{g} \in A(Y)$,

$$f_\alpha^*(\bar{g}) = g(\alpha(\bar{y}_1), \dots, \alpha(\bar{y}_n)) = \alpha(\bar{g}).$$

Thus $f_\alpha^* = \alpha$. $\square$

This correspondence is the algebraic mechanism behind the slogan that affine geometry is dual to commutative algebra; the board referred to it as “Yoneda’s philosophy.”

Example 2.10. Let $X = \mathbb{A}^1 \subseteq \mathbb{A}^1$ with coordinate x , and let

$$Y = V(y_1) \subseteq \mathbb{A}^2.$$

Although these are different embeddings, their coordinate rings are isomorphic:

$$A(X) = k[x] \cong k[y_1, y_2]/(y_1) = A(Y), \quad x \mapsto \bar{y}_2.$$

Corollary 2.9 therefore gives an isomorphism of the associated ringed spaces. This motivates an embedding-independent definition.

Definition 2.11 (Affine variety: intrinsic form). An *affine variety* is a ringed space isomorphic to $(X, \mathcal{O}_X)$ for some affine variety $X \subseteq \mathbb{A}^n$ in the old sense.

3 Lecture 3: August 13

Last time we learned to regard an affine variety not merely as a topological space, but as a ringed space carrying its sheaf of regular functions. We begin with an example that shows why the sheaf is essential.

3.1 The cusp and normalization

Example 3.1 (The cusp). Let

$$X = V(x_1^2 - x_2^3) \subseteq \mathbb{A}^2$$

and consider

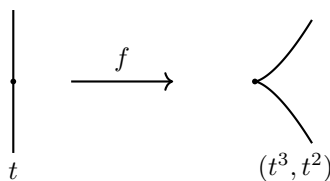
$$f: \mathbb{A}^1 \longrightarrow X, \quad t \longmapsto (t^3, t^2).$$

The map is a morphism, and it is bijective on points. Its pullback on coordinate rings is

$$\begin{aligned} A(X) &\cong k[x_1, x_2]/(x_1^2 - x_2^3) \xrightarrow{f^*} k[t], \\ (\bar{x}_1, \bar{x}_2) &\longmapsto (t^3, t^2). \end{aligned}$$

Thus $A(X)$ is identified with the subring $k[t^2, t^3] \subsetneq k[t]$. In particular, f^* is not an isomorphism, so f is not an isomorphism of affine varieties.

The map can be pictured as follows:



Away from the origin, the set-theoretic inverse is given by

$$(x_1, x_2) \longmapsto \frac{x_1}{x_2}.$$

This expression does not extend to a regular function at the cusp $(0, 0)$. Hence a bijective morphism need not have a regular inverse. The example also shows that the topology alone does not remember the full affine variety: the structure sheaf detects the singularity at the cusp.

This map is the *normalization* of the cusp. Since $\mathbb{A}^1$ is smooth, normalization resolves this singularity; it is therefore a basic one-dimensional example of resolution of singularities.

Proposition 3.2 (The graph trick). Let $X \subseteq \mathbb{A}^n$ be an affine variety and let $f \in A(X)$. Then $D(f)$ is an affine variety, and

$$A(D(f)) \cong A(X)_f.$$

Proof. Define

$$Y = \{(x, t) \in X \times \mathbb{A}^1 : tf(x) = 1\} \subseteq \mathbb{A}^{n+1}.$$

The projection and its inverse are

$$\pi: Y \longrightarrow D(f), \quad (x, t) \longmapsto x, \quad \iota: D(f) \longrightarrow Y, \quad x \longmapsto \left(x, \frac{1}{f(x)}\right).$$

Both maps are morphisms, and they are inverse to one another. Thus $D(f) \cong Y$ is affine. Moreover,

$$A(Y) \cong A(X)[t]/(tf - 1) \cong A(X)_f,$$

where t corresponds to f^{-1} . □

Remark. Introducing an auxiliary coordinate subject to an equation such as $tf = 1$ is a recurring device in algebraic geometry. Similar graph constructions occur, for example, in the study of algebraic groups such as $\mathbb{G}_a$, $\mathbb{G}_m$, and GL_n .

3.2 Prime spectra and Zariski topology

Let R be a commutative ring with identity. Typical examples are $\mathbb{Z}$, $\mathbb{C}[x]$, and $\mathbb{Z}[x]$.

Definition 3.3. The *spectrum* of R is the set

$$\mathrm{Spec} R = \{\mathfrak{p} \subseteq R : \mathfrak{p} \text{ is a prime ideal}\}.$$

We use two notational viewpoints throughout the notes. Capital letters P, Q are used when an element of $\mathrm{Spec} R$ is regarded as a geometric point; lowercase fraktur letters $\mathfrak{p}, \mathfrak{q}$ are used when the same object is regarded as a prime ideal of R . If P corresponds to the prime ideal $\mathfrak{p}_P$, we abbreviate

$$R_P := R_{\mathfrak{p}_P}, \quad \kappa(P) := \mathrm{Frac}(R/\mathfrak{p}_P).$$

The zero ideal (0) is a point of $\mathrm{Spec} R$ precisely when R is an integral domain.

WHAT IS “ $= 0$ ”?

What should the expression

$$“f(P) = 0”$$

mean when the point P is represented by a prime ideal $\mathfrak{p}_P$ rather than by an ordinary k -valued point? Consider the natural maps

$$R \longrightarrow R/\mathfrak{p}_P \longrightarrow \kappa(P) := \mathrm{Frac}(R/\mathfrak{p}_P),$$

where $\kappa(P)$ is the fraction field of the integral domain $R/\mathfrak{p}_P$. Thus the displayed realization of $\kappa(P)$ is a fraction field; standard terminology calls this field the *residue field* at P . Equivalently,

$$\kappa(P) \cong R_{\mathfrak{p}_P}/\mathfrak{p}_P R_{\mathfrak{p}_P} = R_P/\mathfrak{p}_P R_P.$$

We define $f(P)$ to be the image of f in $\kappa(P)$. Then

$$f(P) = 0 \iff f \in \mathfrak{p}_P.$$

Thus “vanishing at the point P ” is translated into the standard algebraic statement that f belongs to the corresponding prime ideal $\mathfrak{p}_P$.

Definition 3.4. For a subset $S \subseteq R$, define its zero locus by

$$\begin{aligned} V(S) &:= \{P \in \mathrm{Spec} R : “f(P) = 0” \text{ for every } f \in S\} \\ &= \{P \in \mathrm{Spec} R : f \in \mathfrak{p}_P \text{ for every } f \in S\} \\ &= \boxed{\{P \in \mathrm{Spec} R : S \subseteq \mathfrak{p}_P\}}. \end{aligned}$$

For a subset $X \subseteq \operatorname{Spec} R$, define

$$\begin{aligned} I(X) &:= \{f \in R : "f(P) = 0" \text{ for every } P \in X\} \\ &= \bigcap_{P \in X} \mathfrak{p}_P. \end{aligned}$$

The *Zariski topology* on $\operatorname{Spec} R$ is the topology whose closed subsets are the sets $V(S)$.

For every ideal $J \subseteq R$ and every closed subset $X \subseteq \operatorname{Spec} R$, one has

$$I(V(J)) = \sqrt{J}, \quad V(I(X)) = X.$$

The first identity follows from the characterization of $\sqrt{J}$ as the intersection of all prime ideals containing J .

Example 3.5. The points of $\operatorname{Spec} \mathbb{Z}$ are

$$\underline{(0)}, (2), (3), (5), (7), \dots$$

Every point (p) associated with a prime number p is closed, whereas (0) is not closed. Indeed, its closure is all of $\operatorname{Spec} \mathbb{Z}$; it is the *generic point* of this irreducible space.

Definition 3.6. For $f \in R$, set

$$D(f) = \operatorname{Spec} R \setminus V(f) = \{P \in \operatorname{Spec} R : f \notin \mathfrak{p}_P\}.$$

We call $D(f)$ a *distinguished open subset* of $\operatorname{Spec} R$.

The distinguished open subsets form a basis for the Zariski topology. In the generality considered here, $D(f)$ need not be an affine variety in the classical sense: R is an arbitrary commutative ring and need not be a finitely generated reduced algebra over a field. Nevertheless, prime ideals of R_f correspond exactly to prime ideals of R that do not contain f , so $D(f)$ is canonically homeomorphic to $\operatorname{Spec} R_f$. Proposition 3.9 will identify its ring of regular functions with R_f .

3.3 The structure sheaf on $\operatorname{Spec} R$

Definition 3.7. Let $U \subseteq \operatorname{Spec} R$ be open. A *regular function* on U is a family

$$\phi = (\phi_P)_{P \in U}$$

such that:

- (1) $\phi_P \in R_P$ for every $P \in U$;
- (2) for every $P \in U$, there exist an open neighbourhood $P \in U_P \subseteq U$ and elements $f, g \in R$ such that, for every $Q \in U_P$,

$$f \notin \mathfrak{p}_Q, \quad \phi_Q = \frac{g}{f} \in R_Q.$$

The ring of regular functions on U is denoted by $\mathcal{O}_{\operatorname{Spec} R}(U)$.

Restriction is obtained by discarding the components outside the smaller open set. These rings and restriction maps form a sheaf, the *structure sheaf* $\mathcal{O}_{\operatorname{Spec} R}$. Thus $(\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R})$ is a ringed space.

Lemma 3.8. For every $P \in \operatorname{Spec} R$, there is a natural isomorphism

$$\mathcal{O}_{\operatorname{Spec} R, P} \cong R_P.$$

Proof. Write $\phi = (\phi_Q)_{Q \in U}$. Evaluation at P defines a ring homomorphism

$$\Theta_P : \mathcal{O}_{\operatorname{Spec} R, P} \longrightarrow R_P, \quad \overline{(U, \phi)} \longmapsto \phi_P.$$

It is well-defined because equivalent representatives agree on a neighbourhood of P . Surjectivity is easy: if $g/f \in R_P$, then $f \notin \mathfrak{p}_P$, and the family

$$\left(\frac{g}{f} \in R_Q \right)_{Q \in D(f)}$$

defines a regular function near P .

For injectivity, suppose $\Theta_P(\overline{(U, \phi)}) = 0$. After replacing U by the neighbourhood U_P from the definition, we may assume that

$$\phi_Q = \frac{g}{f} \in R_Q, \quad f \notin \mathfrak{p}_Q \quad (Q \in U),$$

where $P \in U$. Since $g/f = 0$ in R_P , there exists $h \in R \setminus \mathfrak{p}_P$ such that $hg = 0$ in R . Now $P \in U \cap D(h)$. For every $Q \in U \cap D(h)$, both f and h are invertible in R_Q , and hence $g/f = 0$ in R_Q . Thus ϕ vanishes on the smaller neighbourhood $U \cap D(h)$ of P , so $\overline{(U, \phi)} = 0$ in the stalk. $\square$

Proposition 3.9. *For every $f \in R$, there is a natural isomorphism*

$$\mathcal{O}_{\text{Spec } R}(D(f)) \cong R_f.$$

In particular, $\mathcal{O}_{\text{Spec } R}(\text{Spec } R) \cong R$.

Proof. There is a natural homomorphism

$$\Theta_f: R_f \longrightarrow \mathcal{O}_{\text{Spec } R}(D(f)), \quad \frac{g}{f^m} \longmapsto \left(\frac{g}{f^m} \in R_Q \right)_{Q \in D(f)}.$$

- *Injectivity.* Suppose that $\Theta_f(g/f^m) = 0$, and set

$$J = \{h \in R : hg = 0\}.$$

For every $P \in D(f)$, the equality $g/f^m = 0$ in R_P yields some $h_P \in R \setminus \mathfrak{p}_P$ such that $h_P g = 0$. Hence $h_P \in J \setminus \mathfrak{p}_P$, so $P \notin V(J)$. Therefore

$$V(J) \subseteq V(f).$$

It follows that $f \in \sqrt{J}$. Thus $f^N g = 0$ for some N , which says $g/f^m = 0$ in R_f .

- *Surjectivity.* Let $\phi \in \mathcal{O}_{\text{Spec } R}(D(f))$. For each $P \in D(f)$, shrink a local presentation to a distinguished neighbourhood

$$P \in D(f_P) \subseteq D(f)$$

on which

$$\phi_Q = \frac{g_P}{f_P} \in R_Q \quad (Q \in D(f_P)).$$

Thus

$$D(f) = \bigcup_{P \in D(f)} D(f_P).$$

Equivalently, $V((f_P)_{P \in D(f)}) \subseteq V(f)$, so $f \in \sqrt{(f_P)_{P \in D(f)}}$. Hence finitely many points $P_1, \dots, P_r \in D(f)$ suffice, and for some N_0 and $k_P \in R$,

$$f^{N_0} = \sum_{P \in \{P_1, \dots, P_r\}} k_P f_P.$$

Here and below, P and Q range over the finite set $\{P_1, \dots, P_r\}$. On

$$D(f_P) \cap D(f_Q) = D(f_P f_Q),$$

the sections g_P/f_P and g_Q/f_Q agree. By the injectivity already proved, their images are equal in $R_{f_P f_Q}$. Therefore, for some integer $n_{P,Q} \geq 0$,

$$(f_P f_Q)^{n_{P,Q}} (g_P f_Q - g_Q f_P) = 0 \quad \text{in } R.$$

Because only finitely many pairs occur, one integer n works for every pair. Replace

$$g_P \text{ by } G_P = g_P f_P^n, \quad f_P \text{ by } F_P = f_P^{n+1}.$$

Then $D(F_P) = D(f_P)$, the represented local section is unchanged, and

$$G_P F_Q - G_Q F_P = 0 \quad \text{in } R$$

for all P, Q . Since the $D(F_P)$ still cover $D(f)$, there exist an integer $N \geq 0$ and elements $a_P \in R$ such that

$$f^N = \sum_P a_P F_P.$$

Set $G = \sum_P a_P G_P$. For every Q in the finite index set,

$$G F_Q = \sum_P a_P G_P F_Q = \sum_P a_P G_Q F_P = G_Q f^N.$$

Therefore $G/f^N = G_Q/F_Q$ on $D(F_Q)$. These open sets cover $D(f)$, so $\Theta_f(G/f^N) = \phi$.

Taking $f = 1$ gives $\mathcal{O}_{\text{Spec } R}(\text{Spec } R) \cong R$. □

4 Lecture 4: August 14

Last time we equipped $\text{Spec } R$ with its Zariski topology and its sheaf of regular functions. We now isolate the local property of this ringed space and use it to define schemes and their morphisms.

4.1 Locally ringed spaces and affine schemes

Definition 4.1. A *locally ringed space* is a ringed space $(X, \mathcal{O}_X)$ such that the stalk $\mathcal{O}_{X,P}$ is a local ring for every $P \in X$.

Example 4.2. Affine varieties and the ringed spaces $\text{Spec } R$ are locally ringed spaces. Indeed, Lemma 3.8 gives

$$\mathcal{O}_{\text{Spec } R, P} \cong R_P,$$

and $R_P = R_{\mathfrak{p}_P}$ is a local ring whose unique maximal ideal is $\mathfrak{p}_P R_P$.

Definition 4.3. A *morphism of locally ringed spaces*

$$f: (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

consists of a continuous map $f: X \rightarrow Y$ and ring homomorphisms

$$f_U^*: \mathcal{O}_Y(U) \longrightarrow \mathcal{O}_X(f^{-1}(U))$$

for all open subsets $U \subseteq Y$, satisfying the following conditions.

- (1) The pullbacks are compatible with restriction: if $U \subseteq V$ is open and $\phi \in \mathcal{O}_Y(V)$, then

$$f_U^*(\phi|_U) = f_V^*(\phi)|_{f^{-1}(U)}.$$

- (2) For every $P \in X$, the induced homomorphism on stalks

$$f_P^*: \mathcal{O}_{Y, f(P)} \longrightarrow \mathcal{O}_{X, P}$$

is local. If $\mathfrak{m}_P$ and $\mathfrak{m}_{f(P)}$ denote the respective maximal ideals, this means

$$(f_P^*)^{-1}(\mathfrak{m}_P) = \mathfrak{m}_{f(P)}.$$

The compatibility in (1) is exactly what makes the stalk map well-defined. Explicitly,

$$f_P^*: \overline{(U, \phi)} \mapsto \overline{(f^{-1}(U), f_U^*(\phi))}.$$

Equivalent representatives agree after restriction to a smaller neighbourhood, and their pullbacks therefore do the same.

Remark. For varieties, elements of $\mathcal{O}_Y(U)$ are literally k -valued regular functions. On a general ringed space they are more properly called *sections*. The notation f^* is retained because the same operation is the pullback of functions in the geometric examples; abstractly, each f_U^* is simply a ring homomorphism between rings of sections.

Definition 4.4. An *affine scheme* is a locally ringed space isomorphic to $(\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R})$ for some commutative ring R .

Proposition 4.5 (Affine duality). *For any two commutative rings R and S , pullback gives a natural bijection*

$$\operatorname{Mor}_{\operatorname{LRS}}(\operatorname{Spec} R, \operatorname{Spec} S) \xrightarrow{\sim} \operatorname{Hom}_{\operatorname{Ring}}(S, R), \quad f \mapsto f^*.$$

Thus arrows between affine schemes run in the direction opposite to arrows between their coordinate rings.

Proof. If $f: \operatorname{Spec} R \rightarrow \operatorname{Spec} S$ is a morphism, then its pullback on global sections is a ring homomorphism

$$f^*: S \cong \mathcal{O}_{\operatorname{Spec} S}(\operatorname{Spec} S) \longrightarrow \mathcal{O}_{\operatorname{Spec} R}(\operatorname{Spec} R) \cong R,$$

where Proposition 3.9 was used twice.

Conversely, let $\alpha: S \rightarrow R$ be a ring homomorphism. First define a map of sets

$$f_\alpha: \operatorname{Spec} R \longrightarrow \operatorname{Spec} S, \quad P \mapsto f_\alpha(P), \quad \mathfrak{p}_{f_\alpha(P)} = \alpha^{-1}(\mathfrak{p}_P).$$

The inverse image of a prime ideal under a ring homomorphism is prime, so the map is well-defined. Moreover, for every $s \in S$,

$$f_\alpha^{-1}(D(s)) = D(\alpha(s)),$$

which proves continuity because distinguished opens form a basis.

On a distinguished open $D(s) \subseteq \operatorname{Spec} S$, define the pullback by the localized homomorphism

$$S_s \longrightarrow R_{\alpha(s)}, \quad \frac{a}{s^n} \mapsto \frac{\alpha(a)}{\alpha(s)^n}.$$

These homomorphisms commute with restrictions and therefore determine a morphism of sheaves. At $P \in \operatorname{Spec} R$, the induced map

$$S_{\alpha^{-1}(\mathfrak{p}_P)} \longrightarrow R_{\mathfrak{p}_P} = R_P$$

is local. Hence f_α is a morphism of locally ringed spaces. The two constructions are inverse to one another. $\square$

The proof follows the three layers emphasized in the lecture:

$$\boxed{\text{sets}} \longrightarrow \boxed{\text{topology}} \longrightarrow \boxed{\text{structure sheaves}}.$$

Definition 4.6 (Closed affine subscheme). Let $J \subseteq R$ be an ideal. The quotient homomorphism

$$R \twoheadrightarrow R/J$$

induces a morphism

$$i: \operatorname{Spec}(R/J) \longrightarrow \operatorname{Spec} R.$$

This morphism identifies $\operatorname{Spec}(R/J)$ with a *closed affine subscheme* of $\operatorname{Spec} R$. Its underlying closed subset is $V(J)$.

More generally, a morphism $\text{Spec } S \rightarrow \text{Spec } R$ induced by a surjection $R \twoheadrightarrow S$ has this form with J equal to its kernel. Thus

$$\{\text{closed affine subschemes of } \text{Spec } R\} \longleftrightarrow \{\text{ideals } J \subseteq R\}.$$

The word “closed” is essential here: a surjective homomorphism does not describe an arbitrary affine subscheme.

Example 4.7. Let $R = \mathbb{C}[x]$ and let

$$S_0 = \mathbb{C}[x]/(x) \cong \mathbb{C}, \quad S_1 = \mathbb{C}[x]/(x^2).$$

The spectrum of S_0 consists of a single point Q_0 , corresponding to the prime ideal $(0) \subseteq S_0$. The spectrum of S_1 also consists of a single point Q_1 , but here the corresponding prime ideal is $(\bar{x}) \subseteq S_1$. Under the morphisms induced by the two quotient maps $R \twoheadrightarrow S_i$, both points are sent to the origin $P \in \text{Spec } R$, corresponding to $\mathfrak{p}_P = (x)$.

Thus $\text{Spec } S_0$ and $\text{Spec } S_1$ have the same one-point underlying topological space and the same residue field $\mathbb{C}$, but they are not the same scheme. The ring S_0 is reduced, whereas S_1 contains the nonzero nilpotent element $\bar{x}$, with

$$\mathfrak{m}_{Q_1} = (\bar{x}), \quad \mathfrak{m}_{Q_1}^2 = 0.$$

Geometrically, $\text{Spec } S_1$ is a nonreduced *double point*, or a first-order infinitesimal thickening of the reduced point $\text{Spec } S_0$. Its extra “thickness” is carried by the structure sheaf, not by an additional topological point; equivalently, S_1 has length 2 over $\mathbb{C}$ and $\mathfrak{m}_{Q_1}/\mathfrak{m}_{Q_1}^2 \cong \mathbb{C}$, while S_0 has length 1 and $\mathfrak{m}_{Q_0} = 0$.

Proposition 4.8. *Let R be a ring and let $f \in R$. Then the homomorphism $R \rightarrow R_f$ induces an isomorphism of locally ringed spaces*

$$\text{Spec } R_f \xrightarrow{\sim} D(f) \subseteq \text{Spec } R.$$

In particular, $D(f)$ is an affine scheme.

Proof. Let $Q \in \text{Spec } R_f$ be the point corresponding to a prime ideal $\mathfrak{q}_Q \subseteq R_f$. The map sends Q to the point $P \in \text{Spec } R$ corresponding to the contracted prime ideal $\mathfrak{p}_P = \mathfrak{q}_Q \cap R$. This gives a bijection from $\text{Spec } R_f$ onto the points $P \in \text{Spec } R$ for which $f \notin \mathfrak{p}_P$, namely $D(f)$. It is a homeomorphism because corresponding distinguished opens satisfy

$$D_{\text{Spec } R_f}(g/1) \longleftrightarrow D_R(g) \cap D_R(f) = D_R(fg).$$

On these opens the rings of sections agree:

$$(R_f)_{g/1} \cong R_{fg}.$$

These isomorphisms are compatible with restriction, so the structure sheaves agree as well. $\square$

4.2 Schemes, prevarieties, and gluing

Definition 4.9. A *scheme* is a locally ringed space that has an open cover by affine schemes.

Both the scheme condition and the notion of isomorphism are local. For example, a morphism is an isomorphism if it restricts to compatible isomorphisms on open covers of its source and target.

Definition 4.10. Fix the algebraically closed field k used in the previous lectures. A *prevariety* is a ringed space that has a finite open cover by affine varieties over k .

One obtains prevarieties that are not themselves affine by gluing affine varieties along open subsets. Let I be a finite index set, and suppose that the following information is given:

- affine varieties X_i indexed by $i \in I$;
- open subsets $U_{ij} \subseteq X_i$;

- isomorphisms

$$f_{ij}: U_{ij} \xrightarrow{\sim} U_{ji}.$$

Assume the transition maps satisfy the cocycle conditions

$$U_{ii} = X_i, \quad f_{ii} = \text{id}_{X_i}, \quad f_{ji} = f_{ij}^{-1},$$

and, for every i, j, k ,

$$f_{ij}(U_{ij} \cap U_{ik}) = U_{ji} \cap U_{jk}, \quad f_{ik} = f_{jk} \circ f_{ij} \quad \text{on } U_{ij} \cap U_{ik}.$$

Proposition 4.11 (Gluing construction). *The above information determines a prevariety X as follows.*

- As a set,

$$X = \coprod_{i \in I} X_i / \sim, \quad x \sim f_{ij}(x) \quad (x \in U_{ij}).$$

Let $s_i: X_i \rightarrow X$ denote the canonical maps.

- As a topological space, a subset $U \subseteq X$ is open if and only if $s_i^{-1}(U)$ is open in X_i for every i .
- As a ringed space, for every open subset $U \subseteq X$, set

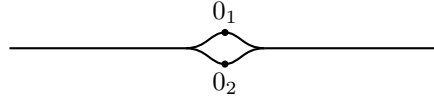
$$\mathcal{O}_X(U) = \{ \phi: U \rightarrow k : s_i^* \phi \in \mathcal{O}_{X_i}(s_i^{-1}(U)) \text{ for every } i \}.$$

The maps s_i identify the X_i with open subprevarieties of X , and the result is unique up to a unique isomorphism compatible with these maps.

Example 4.12 (The affine line with two origins). Take $X_1 = X_2 = \mathbb{A}_k^1$, let

$$U_{12} = X_1 \setminus \{0\}, \quad U_{21} = X_2 \setminus \{0\},$$

and glue by $f_{12}(x) = x$. Every nonzero point in the first copy is identified with the corresponding point in the second, while the two origins 0_1 and 0_2 remain distinct. Schematically, the resulting prevariety looks like



This is a basic example of a nonseparated prevariety; a later separation condition will rule it out from the class of varieties.

Example 4.13 (The projective line). Again take two copies $X_1 = X_2 = \mathbb{A}_k^1$ and delete the origins, but now glue the overlap by

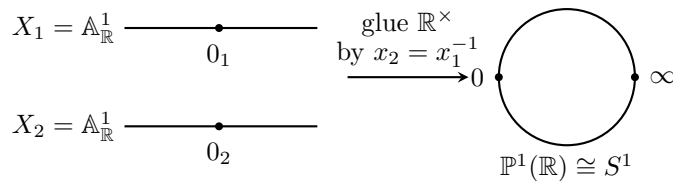
$$f_{12}: U_{12} \longrightarrow U_{21}, \quad x \longmapsto \frac{1}{x}.$$

The resulting prevariety is the projective line $\mathbb{P}_k^1$. When $k = \mathbb{R}$, its real points form a circle,

$$\mathbb{P}^1(\mathbb{R}) \cong S^1;$$

when $k = \mathbb{C}$, its complex points form the Riemann sphere $\mathbb{P}^1(\mathbb{C}) \cong S^2$.

For $k = \mathbb{R}$, the gluing can be pictured as follows. The two origins remain as the points 0 and ∞ , while the two copies of $\mathbb{R}^\times$ are identified by inversion.



4.3 Open and closed subspaces

Remark 4.14 (Open subprevarieties). If X is a prevariety and $U \subseteq X$ is open, then U , with the restricted topology and structure sheaf, is naturally an open subprevariety of X .

Let $Y \subseteq X$ be a closed subset of a prevariety. If $U \subseteq Y$ is open in the subspace topology, let $\mathcal{O}_Y(U)$ consist of the functions $\phi: U \rightarrow k$ that locally extend to regular functions on X : for every $a \in U$, there are an open neighbourhood $a \in V \subseteq X$ and a section $\psi \in \mathcal{O}_X(V)$ such that $\phi = \psi$ on $U \cap V$.

This is a sheaf because restrictions and gluing are inherited from $\mathcal{O}_X$. On every affine open chart $X_i \subseteq X$, the set $Y \cap X_i$ is an affine algebraic set, and this sheaf restricts to its usual sheaf of regular functions. These affine pieces therefore make $(Y, \mathcal{O}_Y)$ a closed subprevariety.

For schemes, the analogous constructions retain ideal-theoretic information.

Definition 4.15. Let X be a scheme.

- (1) If $U \subseteq X$ is open, then $(U, \mathcal{O}_X|_U)$ is the *open subscheme* determined by U .
- (2) A morphism $i: Y \rightarrow X$ is a *closed immersion*, and Y is a *closed subscheme* of X , if there is an affine open cover $X = \bigcup_\lambda V_\lambda$, with $V_\lambda = \text{Spec } R_\lambda$, such that for every λ there is an ideal $J_\lambda \subseteq R_\lambda$ and an identification

$$i^{-1}(V_\lambda) \cong \text{Spec}(R_\lambda/J_\lambda),$$

under which the restricted morphism is induced by the quotient $R_\lambda \twoheadrightarrow R_\lambda/J_\lambda$.

Equivalently, a closed immersion is a homeomorphism from Y onto a closed subset of X for which the sheaf homomorphism

$$\mathcal{O}_X \longrightarrow i_*\mathcal{O}_Y$$

is surjective. Unlike an open subscheme, a closed subscheme is not determined by its underlying closed set: ideals with the same radical give the same topological support but may give different nilpotent scheme structures.

5 Lecture 5: August 18

Last week we passed from affine varieties to ringed spaces and affine schemes. We now return to general prevarieties, impose a separation condition, and then construct projective varieties.

5.1 Separation and products

For an ordinary topological space X , the Hausdorff property can be detected by its diagonal:

$$X \text{ is Hausdorff} \iff \Delta_X = \{(x, x) : x \in X\} \text{ is closed in } X \times X,$$

where the product on the right carries the usual product topology. This suggests the algebro-geometric separation condition below.

Definition 5.1. A prevariety X is called a *variety*, or a *separated prevariety*, if the diagonal

$$\Delta_X = \{(x, x) : x \in X\} \subseteq X \times X$$

is closed in the product prevariety $X \times X$.

Here the product is a product in the category of prevarieties, rather than merely a product of the underlying topological spaces.

Definition 5.2 (Product of prevarieties). Let X and Y be prevarieties. A *product* of X and Y is a prevariety P , together with morphisms

$$\pi_X: P \longrightarrow X, \quad \pi_Y: P \longrightarrow Y,$$

with the following universal property: for every prevariety Z and every pair of morphisms $f_X: Z \rightarrow X$ and $f_Y: Z \rightarrow Y$, there is a unique morphism $f: Z \rightarrow P$ for which

$$\pi_X \circ f = f_X, \quad \pi_Y \circ f = f_Y.$$

Equivalently, the following diagram commutes.

$$\begin{array}{ccccc} & & Z & & \\ & f_X \swarrow & \vdots \downarrow \exists! f & \searrow f_Y & \\ X & \xleftarrow{\pi_X} & P & \xrightarrow{\pi_Y} & Y \end{array}$$

We then write $P = X \times Y$.

Proposition 5.3. *The product $X \times Y$ of two prevarieties exists and is unique up to a unique isomorphism compatible with the two projections.*

Sketch of construction. If $X \subseteq \mathbb{A}^m$ and $Y \subseteq \mathbb{A}^n$ are affine varieties, take the usual algebraic set

$$X \times Y \subseteq \mathbb{A}^{m+n}.$$

Its coordinate ring is

$$A(X \times Y) \cong A(X) \otimes_k A(Y),$$

and the coordinate projections have the required universal property.

In general, choose affine open covers $X = \bigcup_i U_i$ and $Y = \bigcup_j V_j$. The affine products $U_i \times V_j$ are glued along

$$(U_i \cap U_{i'}) \times (V_j \cap V_{j'})$$

using the evident transition isomorphisms. The gluing proposition from the previous lecture produces $X \times Y$ and the two projections. Uniqueness is forced by the universal property. $\square$

Remark (The topology is not the product topology). Even though the underlying set is the Cartesian product, its Zariski topology is generally finer than the product topology of the two underlying Zariski spaces. For example,

$$\mathbb{A}^1 \times \mathbb{A}^1 \cong \mathbb{A}^2$$

has the Zariski topology of the affine plane. Its diagonal is the closed set $V(x - y)$. By contrast, the diagonal is not closed in the topological product of two copies of the non-Hausdorff Zariski space $\mathbb{A}^1$. This is why the product prevariety must be constructed algebraically.

Example 5.4 (Affine varieties are separated). Let $X \subseteq \mathbb{A}^n$ be affine. Inside $X \times X \subseteq \mathbb{A}^{2n}$, with coordinates $(x_1, \dots, x_n, y_1, \dots, y_n)$, the diagonal is cut out by

$$x_1 - y_1 = \dots = x_n - y_n = 0.$$

Hence Δ_X is closed, so every affine variety is a variety.

Proposition 5.5. *An open or closed subprevariety of a variety is again a variety.*

Proof. Let $i: Y \hookrightarrow X$ be the inclusion. The product construction gives an induced morphism $i \times i: Y \times Y \rightarrow X \times X$, and the following square commutes:

$$\begin{array}{ccc} Y & \xrightarrow{\Delta_Y} & Y \times Y \\ \downarrow i & & \downarrow i \times i \\ X & \xrightarrow{\Delta_X} & X \times X. \end{array}$$

Consequently,

$$\Delta_Y = (i \times i)^{-1}(\Delta_X).$$

Since Δ_X is closed in $X \times X$ and $i \times i$ is continuous, its inverse image is closed in $Y \times Y$. Thus Y is separated. $\square$

Remark (What the proof uses). The notation $Y \times Y \subseteq X \times X$ is shorthand for the morphism $i \times i$ above. If Y is open (respectively closed) in X , then $Y \times Y$ is an open (respectively closed) subprevariety of $X \times X$. The point is not that the product carries the ordinary product topology—it does not in general—but that the product prevariety supplies the morphism $i \times i$ and its intrinsic Zariski topology. The formula $\Delta_Y = (i \times i)^{-1}(\Delta_X)$ then proves closedness in that intrinsic topology. The hypotheses “open or closed subprevariety” ensure that Y already has the required prevariety structure and that i is the corresponding immersion.

5.2 Projective algebraic sets

Recall that

$$\mathbb{P}^n = (k^{n+1} \setminus \{0\})/k^\times,$$

and write its points as $[x_0 : \cdots : x_n]$. A polynomial equation on $\mathbb{P}^n$ must be homogeneous: if F has degree d , then

$$F(\lambda x_0, \dots, \lambda x_n) = \lambda^d F(x_0, \dots, x_n),$$

so the condition $F(P) = 0$ is independent of the chosen homogeneous coordinates of P .

Definition 5.6. Let $\mathcal{S}$ be a set of homogeneous polynomials in $k[x_0, \dots, x_n]$. Its projective zero locus is

$$V(\mathcal{S}) = \{P \in \mathbb{P}^n : F(P) = 0 \text{ for every } F \in \mathcal{S}\}.$$

A subset of this form is called a *projective algebraic set*, or, in the terminology of the lecture, a *projective variety*. These sets are the closed subsets of the Zariski topology on $\mathbb{P}^n$.

Let $S = k[x_0, \dots, x_n]$, graded by total degree, and put

$$S_+ = (x_0, \dots, x_n),$$

the *irrelevant ideal*.

Definition 5.7. For $X \subseteq \mathbb{P}^n$, its homogeneous defining ideal is

$$I(X) = \left\langle F \in S_d : d \geq 0, F(P) = 0 \text{ for every } P \in X \right\rangle.$$

Thus $I(X)$ is the homogeneous ideal generated by all homogeneous equations vanishing on X .

Conversely, if $J \subseteq S$ is a homogeneous ideal, then

$$V(J) = \{P \in \mathbb{P}^n : F(P) = 0 \text{ for every homogeneous } F \in J\}.$$

Example 5.8. The extreme cases are

$$V((1)) = \emptyset, \quad V((0)) = \mathbb{P}^n, \quad V(S_+) = V(x_0, \dots, x_n) = \emptyset.$$

The last equality is a genuinely projective phenomenon: all coordinates can vanish only at the origin of $\mathbb{A}^{n+1}$, and the origin does not represent a point of $\mathbb{P}^n$.

Remark (Projective Nullstellensatz). For every projective algebraic set X ,

$$\boxed{V(I(X)) = X}.$$

If J is homogeneous and $V(J) \neq \emptyset$, then

$$\boxed{I(V(J)) = \sqrt{J}}.$$

The nonemptiness qualification matters: taking $J = S_+$ gives $V(J) = \emptyset$, so $I(V(J)) = S$, whereas $\sqrt{J} = S_+$. A formulation valid for every homogeneous ideal is

$$I(V(J)) = \sqrt{J : S_+^\infty}, \quad J : S_+^\infty = \bigcup_{r \geq 0} (J : S_+^r).$$

Thus projective zero loci remember the saturated homogeneous ideal, rather than an irrelevant component supported only at the origin of the affine cone.

Definition 5.9. For a projective variety $X \subseteq \mathbb{P}^n$, the graded ring

$$S(X) = k[x_0, \dots, x_n]/I(X)$$

is called the *homogeneous coordinate ring* of X .

Unlike the affine coordinate ring, $S(X)$ is not in general the ring of global regular functions on X ; its grading records the projective embedding.

5.3 Projective charts and regular functions

For $0 \leq i \leq n$, set

$$U_i = \{[x_0 : \dots : x_n] \in \mathbb{P}^n : x_i \neq 0\}.$$

Then $\mathbb{P}^n = \bigcup_{i=0}^n U_i$, and division by x_i gives an isomorphism $U_i \cong \mathbb{A}^n$. For example,

$$\theta_0: U_0 \xrightarrow{\sim} \mathbb{A}^n, \quad [x_0 : x_1 : \dots : x_n] \mapsto \left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0} \right).$$

Its inverse sends $(a_1, \dots, a_n)$ to $[1 : a_1 : \dots : a_n]$. Under this identification, the induced Zariski topology and the sheaf of regular functions on U_0 are exactly the usual affine ones. Thus one may view

$$\mathbb{A}^n \cong \mathbb{P}^n \setminus V(x_0)$$

as the standard affine chart of projective space.

If $f \in k[t_1, \dots, t_n]$ has degree d , its homogenization is

$$f^h(x_0, \dots, x_n) = x_0^d f\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

This is a homogeneous polynomial of degree d , and dehomogenizing by putting $x_0 = 1$ recovers f . For instance,

$$f(t) = t^2 + 1 \quad \mapsto \quad f^h(x_0, x_1) = x_1^2 + x_0^2.$$

Homogenization adds the possible points at infinity and produces the projective closure of an affine algebraic set; for an ideal, one homogenizes the ideal and then removes irrelevant components by saturation.

Definition 5.10. Let $X \subseteq \mathbb{P}^n$ be a projective variety and let $U \subseteq X$ be open. A map $\phi: U \rightarrow k$ is a *regular function* if, for every $a \in U$, there are an open neighbourhood $U_a \subseteq U$, an integer $d \geq 0$, and homogeneous elements $f, g \in S(X)_d$ of the same degree such that f has no zero on U_a and

$$\phi(P) = \frac{g(P)}{f(P)} \quad (P \in U_a).$$

The ring of regular functions on U is denoted by $\mathcal{O}_X(U)$.

The equal-degree condition makes the quotient independent of homogeneous coordinates, since

$$\frac{g(\lambda x)}{f(\lambda x)} = \frac{\lambda^d g(x)}{\lambda^d f(x)} = \frac{g(x)}{f(x)}.$$

The assignments $U \mapsto \mathcal{O}_X(U)$ form a sheaf, so a projective variety is naturally a ringed space. Moreover, each $X \cap U_i$ is an affine variety, obtained by dehomogenizing the equations of X with respect to x_i . Consequently every projective variety is a prevariety.

5.4 Separation of projective varieties

It remains to check the diagonal condition. A point of $\mathbb{P}^n \times \mathbb{P}^n$ is written

$$([x_0 : \dots : x_n], [y_0 : \dots : y_n]).$$

The two projective points are equal precisely when their representative vectors are proportional, equivalently when

$$x_i y_j - x_j y_i = 0 \quad (0 \leq i, j \leq n).$$

These equations are bihomogeneous, and therefore define a closed subset of $\mathbb{P}^n \times \mathbb{P}^n$. Hence

$$\Delta_{\mathbb{P}^n} = V(x_i y_j - x_j y_i : 0 \leq i, j \leq n)$$

is closed.

On a standard chart the same argument becomes the usual affine diagonal. For every i , the following square commutes:

$$\begin{array}{ccc} U_i & \xhookrightarrow{\Delta} & U_i \times U_i \\ \theta_i \downarrow \sim & & \sim \downarrow \theta_i \times \theta_i \\ \mathbb{A}^n & \xhookrightarrow{u \mapsto (u, u)} & \mathbb{A}^n \times \mathbb{A}^n. \end{array}$$

For $\mathbb{P}^1$, the four product charts also make the gluing explicit. Write U_0, U_1 for the standard charts of the first factor and V_0, V_1 for those of the second factor. Thus $\mathbb{P}^1 \times \mathbb{P}^1$ is obtained by gluing the four affine planes $U_i \times V_j \cong \mathbb{A}^2$. To describe the diagonal map itself, use a third notation for the source:

$$W_0 = \{[x_0 : x_1] : x_0 \neq 0\}, \quad W_1 = \{[x_0 : x_1] : x_1 \neq 0\}.$$

Let $t = x_1/x_0$ on W_0 and $s = x_0/x_1$ on W_1 . Then

$$\begin{aligned} \Delta|_{W_0} : W_0 &\longrightarrow U_0 \times V_0 \cong \mathbb{A}^2, & t &\longmapsto (t, t), \\ \Delta|_{W_1} : W_1 &\longrightarrow U_1 \times V_1 \cong \mathbb{A}^2, & s &\longmapsto (s, s). \end{aligned}$$

On the overlap $W_0 \cap W_1 \cong \mathbb{G}_m$, the same point may instead be viewed in the mixed chart $U_0 \times V_1$; there

$$t \longmapsto (t, t^{-1}),$$

because the second coordinate is $s = y_0/y_1 = t^{-1}$. This is exactly the transition dictated by the gluing. Accordingly, the diagonal is cut out on the four charts by

chart	equation of the diagonal
$U_0 \times U_0$	$t_1 - t_2 = 0$
$U_1 \times U_1$	$s_1 - s_2 = 0$
$U_0 \times U_1$	$t_1 s_2 - 1 = 0$
$U_1 \times U_0$	$s_1 t_2 - 1 = 0$

where in a mixed chart the subscript indicates the factor. Hence the local descriptions agree on all intersections and define a closed subprevariety of the glued product.

Proposition 5.11. *Projective space $\mathbb{P}^n$ is a variety. Consequently every projective variety $X \subseteq \mathbb{P}^n$ is a variety.*

Proof. The equations above show that $\Delta_{\mathbb{P}^n}$ is closed, so $\mathbb{P}^n$ is separated. A projective variety is a closed subprevariety of $\mathbb{P}^n$, and closed subprevarieties of varieties are separated. $\square$

Remark (Why the line with two origins fails). For the line with two origins from the previous lecture, the diagonal is not closed. If the two origins are denoted by 0_1 and 0_2 , then $(0_1, 0_2)$ does not lie on the diagonal, but every open neighbourhood of this pair meets the diagonal in nonzero points. Thus $(0_1, 0_2)$ lies in the closure of the diagonal. By contrast, inversion in the gluing construction of $\mathbb{P}^1$ produces the cross-chart equation $ts = 1$, which is closed.

Remark (Affine versus projective global functions). For an affine variety X , one has

$$\mathcal{O}_X(X) \cong A(X).$$

Already $\mathbb{P}^1$ shows how different the projective case is. With the same cover $\mathbb{P}^1 = W_0 \cup W_1$ as above, let $h \in \mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$. Since $W_0 \cong \mathbb{A}^1$ and $W_1 \cong \mathbb{A}^1$, there are polynomials

$$p(t) \in k[t], \quad q(s) \in k[s]$$

such that $h|_{W_0} = p(t)$ and $h|_{W_1} = q(s)$. On $W_0 \cap W_1 \cong \mathbb{G}_m$, where $s = t^{-1}$, the gluing condition says

$$p(t) = q(t^{-1}) \quad \text{in } k[t, t^{-1}].$$

But

$$k[t] \cap k[t^{-1}] = k \quad \text{inside } k[t, t^{-1}],$$

so p and q are constant. Therefore

$$\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = k.$$

Over $k = \mathbb{C}$, this is the algebraic version of the familiar complex analytic fact that every holomorphic function on a compact connected Riemann surface is constant. In particular, $\mathbb{P}^1(\mathbb{C})$ is the Riemann sphere; regularity on W_1 means that the polynomial $p(t)$ has no pole at infinity. This is the compact-Riemann-surface counterpart of Liouville's theorem: compactness makes a holomorphic function bounded, and Liouville (or the maximum-modulus principle) then forces it to be constant.

For an irreducible projective variety over an algebraically closed field, a later completeness argument similarly gives $\mathcal{O}_X(X) = k$. More generally, on a reduced projective algebraic set a global regular function is constant on each connected component. Thus the homogeneous coordinate ring $S(X)$ should not be confused with $\mathcal{O}_X(X)$; it records the chosen projective embedding, while global regular functions on a projective variety are far more rigid.

6 Lecture 6: August 19

6.1 Projective morphisms and sheaves

Last time we saw that $\mathcal{O}_{\mathbb{P}^n}(\mathbb{P}^n) = k$, in sharp contrast with affine varieties. Nevertheless, there are many nontrivial projective morphisms; for example,

$$q: \mathbb{P}_{\mathbb{C}}^1 \longrightarrow \mathbb{P}_{\mathbb{C}}^1, \quad [x_0 : x_1] \longmapsto [x_0^2 : x_1^2].$$

Replacing (x_0, x_1) by $(\lambda x_0, \lambda x_1)$ multiplies both target coordinates by λ^2 , so the map is well-defined; on the affine charts it is $t \mapsto t^2$. But its pullback on global regular functions is only $\mathbb{C} \rightarrow \mathbb{C}$, just as for the identity. Global functions alone therefore do not remember projective morphisms: we need sheaves.

Let X be a variety or a scheme.

Definition 6.1. A *sheaf (or presheaf) of $\mathcal{O}_X$ -modules* is a sheaf (or presheaf) $\mathcal{F}$ such that $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module for every open subset $U \subseteq X$, and the module structures are compatible with restriction. Thus, if $U \subseteq V$, $\phi, \psi \in \mathcal{F}(V)$, and $\lambda \in \mathcal{O}_X(V)$, then

$$(\phi - \psi)|_U = \phi|_U - \psi|_U, \quad (\lambda\phi)|_U = \lambda|_U \phi|_U.$$

Definition 6.2. Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves of $\mathcal{O}_X$ -modules. A *morphism of $\mathcal{O}_X$ -modules*

$$f: \mathcal{F} \longrightarrow \mathcal{G}$$

is given by the data of $\mathcal{O}_X(U)$ -module homomorphisms

$$f_U: \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

compatible with restriction. Equivalently, for $U \subseteq V$ and $\phi \in \mathcal{F}(V)$,

$$f_V(\phi)|_U = f_U(\phi|_U).$$

The corresponding square is

$$\begin{array}{ccc} \mathcal{F}(V) & \xrightarrow{f_V} & \mathcal{G}(V) \\ \text{res}_{V,U} \downarrow & & \downarrow \text{res}_{V,U} \\ \mathcal{F}(U) & \xrightarrow{f_U} & \mathcal{G}(U). \end{array}$$

It induces an $\mathcal{O}_{X,P}$ -linear map on every stalk,

$$f_P: \mathcal{F}_P \longrightarrow \mathcal{G}_P.$$

Example 6.3. The structure sheaf $\mathcal{O}_X$ is naturally a sheaf of $\mathcal{O}_X$ -modules over itself. If $\mathcal{F}$ and $\mathcal{G}$ are sheaves of $\mathcal{O}_X$ -modules, then

$$(\mathcal{F} \oplus \mathcal{G})(U) = \mathcal{F}(U) \oplus \mathcal{G}(U)$$

with componentwise restrictions is again a sheaf of $\mathcal{O}_X$ -modules. This gives a new sheaf.

Proposition 6.4. *Let $f: \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves of $\mathcal{O}_X$ -modules. Then f is an isomorphism if and only if*

$$f_P: \mathcal{F}_P \xrightarrow{\sim} \mathcal{G}_P$$

is an isomorphism for every $P \in X$.

Proof. The forward implication is immediate. Conversely, suppose every f_P is an isomorphism.

• *Injectivity.* Let $\phi \in \mathcal{F}(U)$ satisfy $f_U(\phi) = 0$. For every $P \in U$,

$$f_P(\phi_P) = (f_U(\phi))_P = 0.$$

Since f_P is injective, $\phi_P = 0$. Hence there is a neighbourhood $P \in U_P \subseteq U$ such that $\phi|_{U_P} = 0$. Since $\mathcal{F}$ is a sheaf and the U_P cover U , we obtain $\phi = 0$. Thus f_U is injective.

• *Surjectivity.* Let $\psi \in \mathcal{G}(U)$. Since f_P is surjective, for every $P \in U$ there are a neighbourhood $P \in U_P \subseteq U$ and a section $\phi_P \in \mathcal{F}(U_P)$ such that

$$\overline{(U_P, f_{U_P}(\phi_P))} = \overline{(U, \psi)} \in \mathcal{G}_P.$$

After shrinking U_P , we may assume

$$f_{U_P}(\phi_P) = \psi|_{U_P}.$$

For $P, Q \in U$, the restrictions of ϕ_P and ϕ_Q to $U_P \cap U_Q$ have the same image under $f_{U_P \cap U_Q}$. By the injectivity already proved, they are equal. Hence there exists $\phi \in \mathcal{F}(U)$ such that

$$\phi|_{U_P} = \phi_P \quad \text{for every } P \in U.$$

Moreover,

$$f_U(\phi)|_{U_P} = f_{U_P}(\phi_P) = \psi|_{U_P}.$$

Since the U_P cover U and $\mathcal{G}$ is a sheaf, $f_U(\phi) = \psi$. Thus f_U is surjective. □

6.2 Direct images and skyscraper sheaves

Let $f: X \rightarrow Y$ be a morphism of varieties or schemes, and let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules.

Definition 6.5. The *direct image sheaf* $f_*\mathcal{F}$ on Y is defined by

$$(f_*\mathcal{F})(U) = \mathcal{F}(f^{-1}(U)) \quad (U \subseteq Y \text{ open}).$$

Its restriction maps are those of $\mathcal{F}$.

The morphism f supplies homomorphisms

$$f_U^*: \mathcal{O}_Y(U) \longrightarrow \mathcal{O}_X(f^{-1}(U)).$$

They define the $\mathcal{O}_Y(U)$ -module structure on $(f_*\mathcal{F})(U)$ by

$$\lambda \cdot \phi = f_U^*(\lambda)\phi, \quad \lambda \in \mathcal{O}_Y(U), \quad \phi \in (f_*\mathcal{F})(U).$$

Thus $f_*\mathcal{F}$ is a sheaf of $\mathcal{O}_Y$ -modules. In particular, the structure-sheaf part of a morphism f is

$$\boxed{\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X}.$$

Example 6.6 (Skyscraper sheaf). Let Y be a variety over k , let $X = \{P\} \subseteq Y$ be a point, and let $i: X \hookrightarrow Y$ be the inclusion. Then

$$(i_*\mathcal{O}_X)(U) = \begin{cases} k, & P \in U, \\ 0, & P \notin U. \end{cases}$$

This is the *skyscraper sheaf* at P .

6.3 Affine module sheaves and kernels

Let R be a ring, let M be an R -module, and put $X = \operatorname{Spec} R$. Recall that a point $P \in X$ corresponds to the prime ideal $\mathfrak{p}_P$, and write

$$M_P := M_{\mathfrak{p}_P}.$$

Definition 6.7. For an open subset $U \subseteq X$, let $\widetilde{M}(U)$ be the set of families

$$\phi = (\phi_P)_{P \in U}, \quad \phi_P \in M_P,$$

which are locally represented by a single fraction: for every $P \in U$, there are an open neighbourhood $P \in U_P \subseteq U$, an element $g \in M$, and an element $f \in R$ such that $U_P \subseteq D(f)$ and

$$\phi_Q = \frac{g}{f} \in M_Q \quad \text{for every } Q \in U_P.$$

With the evident restrictions and pointwise operations, $\widetilde{M}$ is a sheaf of $\mathcal{O}_X$ -modules.

Example 6.8. • The R -module R gives $\widetilde{R} = \mathcal{O}_X$.

- For every $P \in X$,

$$(\widetilde{M})_P \cong M_P.$$

- For $f \in R$,

$$\widetilde{M}(D(f)) \cong M_f.$$

Taking $f = 1$ gives $\widetilde{M}(X) \cong M$.

Remark (Quasi-coherent sheaves). Sheaves of $\mathcal{O}_X$ -modules of the form $\widetilde{M}$ are called *quasi-coherent sheaves*.

Example 6.9 (Restriction of scalars and direct image). Let $\alpha: R \rightarrow S$ be a ring homomorphism. Put

$$X = \operatorname{Spec} S, \quad Y = \operatorname{Spec} R,$$

and let

$$f: X \longrightarrow Y$$

be the induced morphism. If M is an S -module, then $\widetilde{M}$ is a sheaf of $\mathcal{O}_X$ -modules, while $f_*\widetilde{M}$ is a sheaf of $\mathcal{O}_Y$ -modules. View M as an R -module through α , that is,

$$r \cdot m = \alpha(r)m,$$

and denote this R -module by ${}_R M$. Then

$$\boxed{f_*\widetilde{M} \cong \widetilde{{}_R M}}.$$

Indeed, for every $D(r) \subseteq Y$,

$$(f_* \widetilde{M})(D(r)) = \widetilde{M}(D(\alpha(r))) \cong M_{\alpha(r)} = ({}_R M)_r.$$

Thus restriction of scalars forgets only the action of those scalars in S which do not come from R ; the underlying elements and additive group of M do not change. For example, if $R = k$, $S = k[x]$, and $M = S$, then on $X = \operatorname{Spec} k[x]$ we regard $k[x]$ as a free $k[x]$ -module of rank one, whereas after pushing forward to $Y = \operatorname{Spec} k$ we regard the same $k[x]$ as an infinite-dimensional k -vector space.

Example 6.10 (The kernel sheaf). Let $f: \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves of $\mathcal{O}_X$ -modules. Define

$$(\operatorname{Ker} f)(U) = \operatorname{Ker}(f_U: \mathcal{F}(U) \rightarrow \mathcal{G}(U)).$$

In fact, $\operatorname{Ker} f$ is a sheaf. If $U = \bigcup_i U_i$ and compatible sections $\phi_i \in (\operatorname{Ker} f)(U_i)$ are given, the sheaf property of $\mathcal{F}$ glues them to $\phi \in \mathcal{F}(U)$. Moreover,

$$f_U(\phi)|_{U_i} = f_{U_i}(\phi_i) = 0$$

for every i . Since $\mathcal{G}$ is a sheaf, $f_U(\phi) = 0$, and hence $\phi \in (\operatorname{Ker} f)(U)$.

By contrast, the presheaf

$$U \mapsto \operatorname{Im}(f_U: \mathcal{F}(U) \rightarrow \mathcal{G}(U))$$

need not be a sheaf and must be sheafified. We carry this out in the next lecture.

7 Lecture 7: August 20

7.1 Image sheaves and twisting sheaves

We first complete the discussion of images from the preceding lecture. Let

$$f: \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of sheaves of $\mathcal{O}_X$ -modules. Its *image presheaf* is defined by

$$(\operatorname{Im}' f)(U) = \operatorname{Im}(f_U: \mathcal{F}(U) \rightarrow \mathcal{G}(U)).$$

In general this presheaf is not a sheaf.

Example 7.1 (Why the presheaf image need not be a sheaf). In the complex-analytic setting, consider the exponential morphism

$$\mathcal{O}_{\mathbb{C}} \longrightarrow \mathcal{O}_{\mathbb{C}}^{\times}, \quad f \mapsto e^f.$$

On $U = \mathbb{C} \setminus \{0\}$, the function $g(z) = z$ has a holomorphic logarithm locally, so it is locally in the image. But it has no global holomorphic logarithm on U , and hence it is not in the image on U . Thus locally representable image sections need not glue inside the image presheaf.

Denote the stalk of the image presheaf at P by $(\operatorname{Im}' f)_P$.

Definition 7.2. The *image sheaf* $\operatorname{Im} f$ is defined by

$$(\operatorname{Im} f)(U) = \left\{ \phi = (\phi_P)_{P \in U} : \begin{array}{l} \phi_P \in (\operatorname{Im}' f)_P, \text{ and for every } P \in U, \\ \text{there are } P \in U_P \subseteq U \text{ and } \psi \in (\operatorname{Im}' f)(U_P) \\ \text{such that } \phi_Q = \psi_Q \text{ for all } Q \in U_P \end{array} \right\}.$$

Equivalently, $\operatorname{Im} f$ is the sheafification of $\operatorname{Im}' f$. With this definition one may say that f is injective when $\operatorname{Ker} f = 0$, and surjective when $\operatorname{Im} f = \mathcal{G}$.

We now turn to the sheaves needed on projective space. Put

$$R = k[x_0, \dots, x_n]$$

with its standard grading. For $P \in \mathbb{P}^n$, let T_P be the set of homogeneous polynomials in R that do not vanish at P . Then $T_P^{-1}R$ is a graded ring.

Definition 7.3. Fix $d \in \mathbb{Z}$. For an open subset $U \subseteq \mathbb{P}^n$, define $\mathcal{O}_{\mathbb{P}^n}(d)(U)$ to be the set of families

$$\phi = (\phi_P)_{P \in U}, \quad \phi_P \in (T_P^{-1}R)_d,$$

that are locally represented by one homogeneous fraction: for every $P \in U$, there are an open neighbourhood $P \in U_P \subseteq U$, an integer m , and homogeneous polynomials

$$g \in R_{d+m}, \quad h \in R_m,$$

such that h does not vanish on U_P and

$$\phi_Q = \frac{g}{h} \in (T_Q^{-1}R)_d \quad \text{for every } Q \in U_P.$$

With the evident restrictions, $\mathcal{O}_{\mathbb{P}^n}(d)$ is a sheaf of $\mathcal{O}_{\mathbb{P}^n}$ -modules. It is called the d -th twisting sheaf.

For $d = 0$ this construction recovers the structure sheaf:

$$\mathcal{O}_{\mathbb{P}^n}(0) = \mathcal{O}_{\mathbb{P}^n}.$$

Proposition 7.4. Let $f \in R$ be homogeneous of degree $e > 0$, and let $D(f) \subseteq \mathbb{P}^n$ be the corresponding distinguished open subset. Then

$$\mathcal{O}_{\mathbb{P}^n}(d)(D(f)) \cong (R_f)_d.$$

Proof. A homogeneous fraction g/f^r with $\deg g - re = d$ determines its family of germs on $D(f)$ and hence a section of $\mathcal{O}_{\mathbb{P}^n}(d)$. This gives the natural map

$$(R_f)_d \longrightarrow \mathcal{O}_{\mathbb{P}^n}(d)(D(f)).$$

The same common-denominator argument as in the affine case shows that this map is both injective and surjective. $\square$

Corollary 7.5. Let $d \in \mathbb{Z}$.

(1) If $d < 0$, then

$$\mathcal{O}_{\mathbb{P}^n}(d)(\mathbb{P}^n) = 0.$$

(2) If $d \geq 0$, then

$$\mathcal{O}_{\mathbb{P}^n}(d)(\mathbb{P}^n) \cong R_d.$$

Remark (Recall). For $d = 0$, part (2) gives $\mathcal{O}_{\mathbb{P}^n}(\mathbb{P}^n) = k$. The corresponding statement for a projective curve F is $\mathcal{O}_F(F) = k$: the only rational functions that are regular everywhere on F are constants. See Andreas Gathmann, *Plane Algebraic Curves* (2023), Corollary 6.29.

Example 7.6 (The standard cover of $\mathbb{P}^1$). Let $\mathbb{P}^1 = W_0 \cup W_1$, with $t = x_1/x_0$ on W_0 and $s = x_0/x_1$ on W_1 . Using the local generators x_0^d and x_1^d , a global section of $\mathcal{O}_{\mathbb{P}^1}(d)$ is given by polynomials $p(t)$ and $q(s)$ satisfying

$$p(t) = t^d q(t^{-1}) \quad \text{in } k[t, t^{-1}].$$

If $d < 0$, this forces $p = q = 0$. If $d \geq 0$, the solutions are

$$p(t) = a_0 + a_1 t + \cdots + a_d t^d, \quad q(s) = a_d + a_{d-1} s + \cdots + a_0 s^d.$$

Thus $\mathcal{O}_{\mathbb{P}^1}(d)(\mathbb{P}^1) \cong k[x_0, x_1]_d$. For $d = 0$, this is the earlier computation $\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = k$.

Remark (Tensor products of twists). Multiplication of homogeneous fractions gives a natural isomorphism

$$\mathcal{O}_{\mathbb{P}^n}(a) \otimes_{\mathcal{O}_{\mathbb{P}^n}} \mathcal{O}_{\mathbb{P}^n}(b) \cong \mathcal{O}_{\mathbb{P}^n}(a + b).$$

Thus tensor product of twisting sheaves corresponds to addition of their indices.

Example 7.7 (The sheaf associated with a divisor). Let $F \subseteq \mathbb{P}^2$ be a smooth plane curve and let D be a divisor on F . For an open subset $U \subseteq F$, set

$$\mathcal{O}_F(D)(U) = \{0\} \cup \{\phi \in k(F)^\times : \operatorname{div}(\phi) + D \geq 0 \text{ on } U\}.$$

This is a sheaf of $\mathcal{O}_F$ -modules. To extract numerical invariants from such sheaves, we introduce cohomology.

7.2 Čech cohomology

Let X be a variety over k , let $\mathcal{F}$ be a sheaf on X , and let

$$U_1, \dots, U_r$$

be an affine open cover of X . For $p \in \mathbb{N}$, define

$$C^p(\mathcal{F}) = \bigoplus_{i_0 < \dots < i_p} \mathcal{F}(U_{i_0} \cap \dots \cap U_{i_p}).$$

The boundary map

$$d^p: C^p(\mathcal{F}) \longrightarrow C^{p+1}(\mathcal{F})$$

is given by

$$(d^p \phi)_{i_0 \dots i_{p+1}} = \sum_{j=0}^{p+1} (-1)^j \phi_{i_0 \dots \widehat{i_j} \dots i_{p+1}} \Big|_{U_{i_0} \cap \dots \cap U_{i_{p+1}}}.$$

Lemma 7.8. *For every $p \geq 0$, the map*

$$d^{p+1} \circ d^p: C^p(\mathcal{F}) \longrightarrow C^{p+2}(\mathcal{F})$$

is the zero map.

Proof. After expanding $d^{p+1}(d^p \phi)$, each restriction obtained by deleting two indices occurs twice, with opposite signs. Hence all terms cancel. $\square$

Consequently there is a complex

$$C^0(\mathcal{F}) \xrightarrow{d^0} C^1(\mathcal{F}) \xrightarrow{d^1} C^2(\mathcal{F}) \longrightarrow \dots$$

Definition 7.9. The p -th cohomology of $\mathcal{F}$ is

$$H^p(X, \mathcal{F}) = \frac{\text{Ker } d^p}{\text{Im } d^{p-1}}.$$

We use the convention $C^{-1}(\mathcal{F}) = 0$ and $d^{-1} = 0$, so $H^0(X, \mathcal{F}) = \text{Ker } d^0$.

Remark. Up to isomorphism, the above definition does not depend on the choice of affine open cover.

Lemma 7.10. *Let $\mathcal{F}$ be a sheaf on a variety X . Then:*

- (1) $H^0(X, \mathcal{F}) \cong \mathcal{F}(X)$;
- (2) if X is affine, then $H^p(X, \mathcal{F}) = 0$ for every $p > 0$;
- (3) if $i: X \hookrightarrow Y$ is the inclusion of a closed subvariety, then

$$H^p(Y, i_* \mathcal{F}) \cong H^p(X, \mathcal{F})$$

for every $p \geq 0$.

Proof. The first assertion is exactly the sheaf gluing condition. For the second, use the one-member affine cover $\{X\}$. For the third, choose an affine open cover $Y = \bigcup_i U_i$. Then the nonempty sets $X \cap U_i$ form an affine open cover of X , and

$$(i_* \mathcal{F})(U_{i_0} \cap \dots \cap U_{i_p}) = \mathcal{F}(X \cap U_{i_0} \cap \dots \cap U_{i_p}).$$

The two Čech complexes are therefore identical. $\square$

Proposition 7.11 (Long exact sequence). *Let*

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$$

be a short exact sequence. Then there is a long exact sequence

$$\begin{aligned} 0 \longrightarrow H^0(X, \mathcal{F}_1) \longrightarrow H^0(X, \mathcal{F}_2) \longrightarrow H^0(X, \mathcal{F}_3) \\ \longrightarrow H^1(X, \mathcal{F}_1) \longrightarrow H^1(X, \mathcal{F}_2) \longrightarrow H^1(X, \mathcal{F}_3) \longrightarrow \dots \end{aligned}$$

7.3 Cohomology of projective space

Let

$$\mathcal{F} = \bigoplus_{d \in \mathbb{Z}} \mathcal{O}_{\mathbb{P}^n}(d).$$

Proposition 7.12. *With $R = k[x_0, \dots, x_n]$, one has*

(1)

$$H^0(\mathbb{P}^n, \mathcal{F}) \cong R;$$

(2)

$$H^n(\mathbb{P}^n, \mathcal{F}) \cong \frac{1}{x_0 \cdots x_n} k[x_0^{-1}, \dots, x_n^{-1}];$$

(3)

$$H^q(\mathbb{P}^n, \mathcal{F}) = 0 \quad \text{if } q \neq 0, n.$$

Proof. The first assertion follows from Corollary 7.5. For the remaining assertions, use the standard affine cover

$$U_i = D(x_i) = \{P \in \mathbb{P}^n : x_i(P) \neq 0\}, \quad i = 0, \dots, n.$$

On an intersection of standard charts,

$$\mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_p}) \cong R_{x_{i_0} \cdots x_{i_p}}.$$

To compute the top cohomology, consider the end of the Čech complex:

$$C^{n-1}(\mathcal{F}) \xrightarrow{d^{n-1}} C^n(\mathcal{F}) \longrightarrow C^{n+1}(\mathcal{F}) = 0,$$

where

$$C^{m-1}(\mathcal{F}) = \bigoplus_{i=0}^n R_{x_0 \cdots \widehat{x_i} \cdots x_n}, \quad C^m(\mathcal{F}) = R_{x_0 \cdots x_n}.$$

Hence

$$H^n(\mathbb{P}^n, \mathcal{F}) = \frac{C^n(\mathcal{F})}{\text{Im } d^{n-1}} \cong \frac{R_{x_0 \cdots x_n}}{\sum_{i=0}^n R_{x_0 \cdots \widehat{x_i} \cdots x_n}}.$$

Modulo the denominator, a Laurent monomial survives precisely when every variable has negative exponent. Therefore the quotient is

$$\frac{1}{x_0 \cdots x_n} k[x_0^{-1}, \dots, x_n^{-1}],$$

which proves (2).

For (3), we argue by induction on n using a hyperplane section. Let

$$i: \mathbb{P}^{n-1} = V(x_0) \hookrightarrow \mathbb{P}^n, \quad \mathcal{F}_{n-1} = \bigoplus_{d \in \mathbb{Z}} \mathcal{O}_{\mathbb{P}^{n-1}}(d).$$

For every $d \in \mathbb{Z}$, multiplication by x_0 gives

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n}(d-1) \xrightarrow{x_0} \mathcal{O}_{\mathbb{P}^n}(d) \longrightarrow i_* \mathcal{O}_{\mathbb{P}^{n-1}}(d) \longrightarrow 0.$$

Summing over all twists gives

$$0 \longrightarrow \mathcal{F} \xrightarrow{x_0} \mathcal{F} \longrightarrow i_* \mathcal{F}_{n-1} \longrightarrow 0.$$

By Lemma 7.10, the cohomology of the last term is the cohomology of $\mathcal{F}_{n-1}$ on $\mathbb{P}^{n-1}$. Read the associated long exact sequence degree by degree. By induction, the intermediate cohomology groups on $\mathbb{P}^{n-1}$ vanish; for $q = 1$, also use the surjectivity of the restriction map on global sections, which is immediate from the polynomial description in Corollary 7.5. Thus

$$H^q(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(d-1)) \longrightarrow H^q(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(d))$$

is injective for $0 < q < n$. For $d \gg 0$, the group on the right is zero, as one checks directly on the standard affine cover. Iterating these injections therefore gives

$$H^q(\mathbb{P}^n, \mathcal{F}) = 0 \quad (0 < q < n).$$

This completes the induction; the case $n = 0$ is immediate. $\square$

Example 7.13 (The first cohomology of $\mathbb{P}^2$). We specialize the hyperplane-section argument from Proposition 7.12. Let

$$L = V(x_0) \cong \mathbb{P}^1, \quad i: L \hookrightarrow \mathbb{P}^2.$$

For every $d \in \mathbb{Z}$, multiplication by x_0 gives a short exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(d-1) \xrightarrow{x_0} \mathcal{O}_{\mathbb{P}^2}(d) \longrightarrow i_*\mathcal{O}_L(d) \longrightarrow 0.$$

The relevant part of the associated long exact sequence is

$$\begin{aligned} H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)) &\longrightarrow H^0(L, \mathcal{O}_L(d)) \\ &\longrightarrow H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d-1)) \xrightarrow{x_0} H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)). \end{aligned}$$

The first map is surjective. Indeed, if $d \geq 0$, every homogeneous polynomial of degree d in x_1, x_2 is the restriction of the same polynomial on $\mathbb{P}^2$; if $d < 0$, its target is zero. Exactness therefore shows that

$$H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d-1)) \hookrightarrow H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d))$$

for every d .

It remains to start these injections. For $N \geq 0$, the standard Čech complex shows directly that

$$H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(N)) = 0.$$

Here is the short monomial check. In a homogeneous Laurent monomial of degree N , a term with at least two negative exponents can occur in at most one component of a 1-cochain, so the cocycle relation forces its coefficient to vanish. Every remaining monomial has at most one negative exponent, and the cocycle relation allows its coefficients to be lifted to one of $(R_{x_0})_N, (R_{x_1})_N, (R_{x_2})_N$; hence the cocycle is a coboundary.

Now fix any $d \in \mathbb{Z}$ and choose $N \geq \max\{d, 0\}$. Iterating the injections above gives

$$H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)) \hookrightarrow H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(N)) = 0.$$

Consequently

$$H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)) = 0 \quad \text{for every } d \in \mathbb{Z}.$$